Silence Is Not Violation: Measuring and Reducing False Violation Errors in LLM Compliance Checking of Vendor Data Processing Agreements under India's DPDP Act

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Abstract—Large language models (LLMs) are increasingly used to check commercial contracts against regulations. We identify and measure a specific, recurring failure: a checker that judges each clause in isolation reports a contract's silence on a requirement as a violation of it. We observed this failure while building PactScan, an LLM- and knowledge-graph-based compliance system for vendor Data Processing Agreements (DPAs) under India's Digital Personal Data Protection (DPDP) Act, 2023, where a fully compliant contract received 19 findings. We release a minimal-pair benchmark of 138 synthetic DPAs (1,752 graded contract–requirement pairs) in which each contract follows, omits or breaks exactly one obligation, including "trap" contracts whose amendments hide or cure a violation. We compare the original clause-level checker with a method that (i) instructs the model that silence is neutral, (ii) checks each requirement only against clauses on its subject, (iii) reads each clause together with the provisions that override it, and (iv) decides at the level of the whole contract. On gpt-oss-120b the false violation rate falls from 60.9% to 19.6% while no mandatory gap is missed and 99% of violations are still caught; the same pattern holds on gpt-oss-20b (60.8%→27.1%) and qwen3.8-27b (61.2%→14.1%), and in all 18 of 18 company×model units (sign test $p = 7.6 \times 10^{-6}$), and it holds in each of five independent runs. An ablation shows that the prompt alone barely helps: it shifts the model's verdicts from "contradicts" (65%) to "neutral" (64%), but a clause-level decision rule still turns those neutral verdicts into findings. The fix therefore comes from the decision unit rather than from the prompt. The same holds when existing contracts are re-checked against new regulations: flagging every non-entailing clause flags 100% of compliant contracts, flagging only contradictions misses 96% of gaps, and the contract-level rule misses none (F1 76.5%). Residual false alarms concentrate on compound requirements that bundle several duties into one rule.

Index Terms—legal NLP, contract compliance, large language models, natural language inference, data protection, DPDP Act, GDPR, benchmark

I. Introduction

Companies sign Data Processing Agreements (DPAs) with every vendor that handles personal data on their behalf, and India's Digital Personal Data Protection (DPDP) Act, 2023 [1] makes the company (the Data Fiduciary) accountable for what its vendors (Data Processors) do. Reviewing a portfolio of such contracts against the Act, an internal playbook and, for many firms, the EU GDPR [2] is slow and repetitive, which makes it an obvious target for automation with large language models (LLMs).

A natural design, and the one from which most systems start, is to split the contract into clauses and ask a model, for every clause and every relevant requirement, whether the clause entails, contradicts or is neutral to it, in the style of contract natural language inference [3]. This paper is about a failure of that design that is easy to miss and expensive in practice: a clause that simply does not deal with a requirement is reported as violating it. For example, a cross-border-transfer clause is flagged for not restating breach-notification duties, and a confidentiality clause for never mentioning children's data. Each alarm can be defended in isolation ("this clause does not secure X"), but it is wrong for the contract as a whole, because another clause secures X.

We first encountered this failure during the development of PactScan, a working compliance platform that parses vendor DPAs into a Neo4j knowledge graph and then checks them with an LLM. Its first version raised 19 findings on a contract that we had written to be fully

compliant. Later, a regulation-update feature produced 9 and then 13 findings for a single amendment on a two-page contract. A first fix, which reported only explicit contradictions, stopped the flood but also hid a real problem, namely a contract that never mentions a newly mandatory protection. These incidents, recorded in the project's development log, motivate the study.

Contributions.

1) We characterise the silence-as-violation error and its opposite failure, the missed gap, and define a pair of metrics that measure both.
2) We release a minimal-pair benchmark of 138 synthetic vendor DPAs under the DPDP Act (with GDPR as a second standard): 1,752 graded contract–requirement pairs, including trap contracts in which an amendment hides or cures a violation. To our knowledge it is the first compliance benchmark for DPDP contracts.
3) We evaluate a four-part method (a conservative prompt, subject routing, override-aware reading and a contract-level decision rule) through a step-by-step ablation on three open models, using conservative statistics.
4) We show that the decisive component is the unit of decision, not the prompt, and that residual errors concentrate on compound requirements, which suggests a concrete next step.

II. Related Work

Contract understanding and NLI. CUAD [4] provides expert clause annotations for 510 contracts; we use its category names as a taxonomy reference only. ContractNLI [3] treats contract review as document-level NLI [5] and includes an explicit NotMentioned label. This is the same three-way distinction on which our metrics rely, and in contracts it plays the role that unanswerable questions play in reading comprehension [6]. Legal-understanding tasks are collected in LexGLUE [7] and LegalBench [8], while CLAUDETTE [9] detects potentially unfair clauses in terms of service and Polisis [10] classifies privacy-policy text. CLAUSE [11] introduces ten categories of embedded flaws into CUAD and ContractNLI contracts and reports that LLMs often miss subtle discrepancies and have difficulty justifying them in legal terms. Our benchmark is different in jurisdiction (DPDP) and document type (vendor DPAs), and also in purpose, since we built it to separate silence from violation in matched pairs and not to detect flaws in general.

Compliance checking of DPAs. Amaral et al. [12] and Azeem and Abualhaija [13] check GDPR DPAs for completeness with NLP and machine learning, and Hassani et al. [14] argue that LLM-based compliance analysis should use a broader context than the single sentence. These works establish that completeness (what is missing) is a distinct question from conformity (what is wrong); we study what happens when a checker conflates the two. Work on the DPDP Act so far checks website privacy policies against the Act with access-control rules [15] or embeds DPDP compliance logic in software agents [16]; we found no study of compliance checking for DPDP contracts.

Reliability of LLM legal and compliance verdicts. LLMs hallucinate on legal questions at substantial rates [17], including in commercial legal research tools built on retrieval [18], [19]. For compliance specifically, verdicts are often invariant when the governing rule is deleted, swapped or negated [20], and recent pipelines use NLI only as a screening step before verification by a panel of LLM judges [21]. Regulatory-change monitoring with knowledge-graph-augmented gap detection has been studied for financial institutions' policies [22]. Our study complements this work, since we keep the rule fixed and vary how verdicts are turned into findings.

Minimal pairs. In designing the benchmark we follow the logic of behavioural testing with CheckList [23] and contrast sets [24], which rely on small, controlled edits whose correct effect on the output is known beforehand.

III. The System and the Problem

A. PactScan's Detect pipeline

PactScan parses an uploaded contract, extracts clauses with an LLM into a fixed taxonomy (e.g. Breach Notification, Data Processing, Sub-processor Obligations), and writes them into a Neo4j knowledge graph used for graph-based retrieval [25]. When a clause contains override language ("notwithstanding", "supersedes"), an LLM confirms whether it changes another clause, and an Overrides edge is stored. Detect then checks each clause against the requirements relevant to its type: 8 internal playbook rules and 5 regulation clauses (DPDP Sections 4, 8 and 9; GDPR Articles 6 and 8). Each check is a single batched structured-output call that returns, for every requirement, an Entails/Contradicts/Neutral label together with a confidence and an explanation. The findings then go to human-approved redlining, a later stage that this paper does not touch.

B. Three outcomes, two errors

For a contract c and requirement r, the correct outcome is one of:

- NONE: nothing is wrong, either because some clause satisfies r or because r does not bind c;
- VIOLATES: a clause goes against r;
- GAP: r is mandatory and nothing in c addresses it.

A checker flags (c, r) if it produces at least one finding for r in c. The false violation rate (FVR) is the share of NONE pairs that are flagged. This is the error that the paper addresses. The missed gap rate (MGR) is the share of GAP pairs that are not flagged, which is the kind of error an over-cautious fix creates. We also report violation recall (share of VIOLATES pairs flagged) and precision, recall and F1 over all problems (VIOLATES $\cup$ GAP).

C. How the problem appeared

The original Detect prompt (preserved in the project's history) instructed the model that "a clause that is vague or silent on a specific mandatory protection … CONTRADICTS the requirement rather than being NEUTRAL", and the decision rule turned every contradiction, and every neutral verdict given with high confidence, into a finding. The first live run produced 19 findings on the clean contract. The prompt was rewritten ("merely silent on an adjacent obligation is NEUTRAL, never CONTRADICTS"), routing was narrowed, and a cross-clause check suppressed a children's-data gap when another clause covered it. Months later the regulation-update feature, which re-checks existing contracts against a newly added rule, flagged every non-entailing clause and produced 9, then 13, findings for one amendment; restricting it to explicit contradictions removed the legitimate case of a contract silent on a new mandatory requirement. The final rule, introduced below, decides per contract.

IV. Method

We compose four changes, each addressing one way silence leaks into findings.

(1) Conservative prompt. The model is told that a clause that "does not deal with this requirement's subject" or is "merely silent on an adjacent obligation" is Neutral, and that Contradicts is reserved for a clause that itself creates the problem.

(2) Subject routing. Each requirement is checked only against clauses of the type that governs its subject (e.g. breach notification only against Breach Notification clauses), instead of a broad map that checks, say, every data-processing clause against every regulation.

(3) Override-aware reading. A clause is classified on its effective text: its own text followed by every provision that overrides it, found by traversing Overrides edges up to three hops in the graph.

(4) Contract-level decision. Let $V(c, r)$ be the multiset of verdicts for r over the clauses of c routed to r. The number of findings is

$$f(c,r) = \begin{cases} \#\{\text{Contradicts} \in V\} & \text{if one contradicts } r, \\ 0 & \text{else if one entails } r, \\ 1 & \text{else if } r \text{ mandatory}, \\ 0 & \text{otherwise.} \end{cases} \tag{1}$$

A violation is still reported on the clause that causes it; silence becomes at most one gap finding, and only for a mandatory requirement that nothing in the contract satisfies (Fig. 1).

Whether a requirement is mandatory is a property of the rule set by the playbook author, not of the contract.

Configurations. We add the changes one at a time: A the original checker (original prompt, broad routing, clause-level rule); B A + conservative prompt; C B + subject routing; D C + contract-level decision (the full method). D−ET is D without override-aware reading. P is PactScan's Detect agent in its deployed form, run without modification, and its gap logic is limited to the two children's-data requirements. All configurations use the same batched call, client, temperature and model.

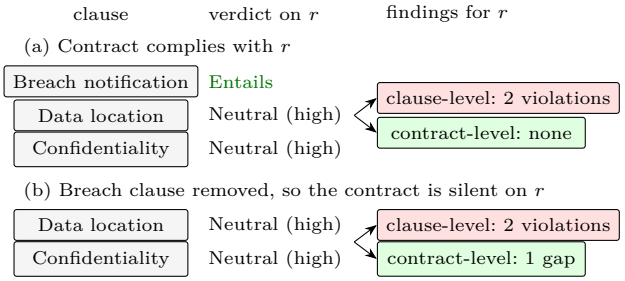


Fig. 1. Two decision rules applied to the same verdicts for a mandatory requirement r ("notify breaches within 72 hours"). In both contracts the clause-level rule reports silence as a violation, whereas the contract-level rule reports nothing in (a) and a single gap in (b).

TABLE I
The benchmark.

Base contracts (one per sector)	6
Contracts	138
Requirements (8 playbook, 3 DPDP, 2 GDPR)	13
Graded (contract, requirement) pairs	1,752
correct answer NONE	1,599
correct answer VIOLATES	96
correct answer GAP	57
Pairs excluded as ambiguous	42
Traps: hidden violation / hidden fix	12 / 12

V. Benchmark

Construction. Six base DPAs pair a fictional Indian company with a vendor in different sectors (health-tech, retail, ed-tech, fintech, HR/payroll, logistics), differing in parties, data categories, locations, notification windows and liability floors, and written in three drafting styles. Each base contains nine topic clauses (breach notification, liability cap, processing details, children's data, sub-processors, data location, deletion on termination, audit rights, lawful basis) and four filler clauses (services, confidentiality, indemnity, governing law), and complies with all 13 requirements. Each base yields 23 contracts, namely the base itself, nine silent variants (one topic clause removed), nine breaks variants (one topic clause rewritten so that it breaks its requirements) and four trap variants. In the traps, which cover breach notification and deletion, a later amendment either hides a violation behind a compliant clause or cures a non-compliant one. Since every variant differs from its base in exactly one place, any change in a checker's output can be attributed to that change.

Labels. Every requirement belongs to a single topic, and its correct answer follows from which topic was changed and in what way. Of the nine topics, seven are

mandatory. The other two are not, because silence on them breaks nothing. A contract with no liability-cap clause has unlimited liability, and the rule only forbids a cap that is too low; a contract that does not state a lawful basis is also acceptable, since having a lawful ground is the Data Fiduciary's own statutory duty. The children's-data rules do not bind the payroll base, which processes only employees' data. The company's playbook requires every other base to address children's data explicitly, so silence on it counts as a gap. Before any model was run, all labels were reviewed clause by clause against the requirement texts. This review led us to rewrite three breaking clauses that also touched another topic's rule and to exclude 42 pairs on which careful readers could disagree (Table I).

Scope of the measurement. Because the checker is given the benchmark's own clauses, clause types and override links, parsing and extraction errors are excluded, and only the verdict step is measured.

VI. Experimental Setup

Models. gpt-oss-120b and gpt-oss-20b [26] and qwen3.8-27b [27], served by GroqCloud. Each run is pinned to a single model, with no fallback to another, and uses temperature 0.2.

Controls. Every run has its own private response cache, which means repeated runs never reuse one another's answers. Within a run, an identical question asked under the same system prompt is answered only once, so each distinct clause text receives exactly one verdict. The system's cache key does not include the system prompt, and for this reason each prompt version is given its own cache. All verdicts are stored together with the model and a timestamp, and runs can be resumed.

Statistics. Confidence intervals (95%) are obtained from a bootstrap [28] over contracts with 2,000 resamples, and per-pair correctness is compared with exact McNemar tests [29]. These figures overstate certainty, since variants of one base share 12 of their 13 clauses and their pairs are therefore not independent. As a more conservative check we also report a sign test that treats each (base, model) combination as a single unit. Agreement across runs and models is reported as the share of pairs answered identically and Fleiss' κ [30].

Experiments. E1 runs all configurations on gpt-oss-120b over all 138 contracts. E2 runs A and C/D on the two other models. E3 runs A and C/D five independent times on two bases (46 contracts); it was run on gpt-oss-20b because the free daily token quota for gpt-oss-120b was exhausted by E1 and E4. E4 re-checks contracts against four simulated DPDP amendments through PactScan's regulation-update path.

VII. Results

A. E1: the error and the ablation

Table II and Fig. 2a show the error in full. The original checker flags 60.9% of the pairs on which nothing is

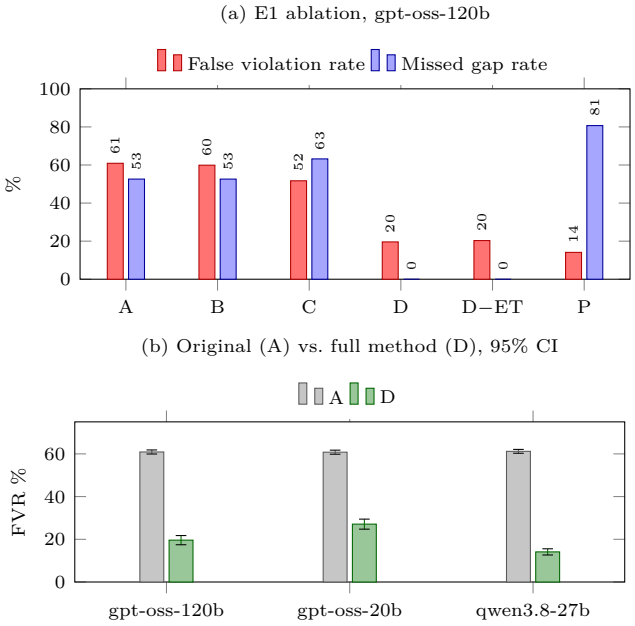


Fig. 2. (a) Adding the four changes one at a time: only the contract-level decision (D) removes most false violations, and it is also the only configuration that misses no gap. (b) The original checker makes the error at the same rate on every model; the full method reduces it on all three.

wrong and raises 21.7 findings on a contract that complies with everything, which is close to the 19 observed during development. It also misses half the gaps: for a removed clause, a mandatory requirement often has no clause of a routed type left to ask about. The full method reduces the false violation rate to 19.6%, misses no gap, keeps violation recall at 99% and raises 2.5 findings on a clean contract. The difference between A and D is significant on matched pairs (691 pairs correct only under D vs. 1 only under A; McNemar $p = 6.7 \times 10^{-206}$) and under the conservative per-base sign test reported in E2.

The prompt is not the fix. Moving from A to B changes the false violation rate by only one point (McNemar $p = 0.004$), although the prompt changes the model's behaviour dramatically (Table III): the share of Contradicts verdicts falls from 65.0% to 7.4% and Neutral rises from 4.0% to 64.4%. Under the clause-level rule, however, a neutral verdict given with high confidence is still turned into a finding, and so silence that the prompt has correctly relabelled ends up being reported anyway. Subject routing (C) removes some of these checks (51.7%), and the contract-level decision (D) removes the rest: 549 pairs become correct and none become incorrect between C and D ($p = 1.1 \times 10^{-165}$).

Effect of override-aware reading on hidden changes. In trap contracts where an amendment hides a violation, D catches 94.4% of the violated pairs, compared with only 27.8% when override-aware reading is removed ($p = 8.0 \times 10^{-7}$ on trap pairs). Where an amendment

TABLE II
E1 on gpt-oss-120b, 138 contracts (%; 95% bootstrap CI in brackets). "Clean" is the mean number of findings on the six base contracts, which are fully compliant.

Configuration	False violation rate	Missed gap rate	Violation recall	F1	Clean
A original checker	60.9 [59.9, 61.8]	52.6 [38.6, 69.8]	100.0	19.7	21.7
B + conservative prompt	59.9 [58.9, 60.9]	52.6 [38.6, 69.8]	100.0	19.9	21.8
C + subject routing	51.7 [50.4, 52.8]	63.2 [46.4, 81.2]	99.0	21.2	12.5
D + contract-level decision (full)	19.6 [17.4, 21.7]	0.0	99.0	49.2	2.5
D−ET full, no override-aware reading	20.3 [18.1, 22.5]	0.0	86.5	45.3	2.5
P PactScan as deployed	14.1 [12.1, 16.1]	80.7 [67.7, 95.2]	77.1	36.6	1.8

TABLE III
Share of raw verdicts by label (gpt-oss-120b, %).

Prompt, routing	Contradicts	Entails	Neutral
A original, broad	65.0	30.9	4.0
B conservative, broad	7.4	28.2	64.4
C/D conservative, by subject	10.1	40.3	49.6

TABLE IV
E2: the original checker (A) and the full method (D) compared on three models (%, 95% CI).

Model		FVR	MGR	Viol. recall
gpt-oss-120b	A	60.9 [59.9, 61.8]	52.6	100.0
	D	19.6 [17.4, 21.7]	0.0	99.0
gpt-oss-20b	A	60.8 [59.8, 61.7]	52.6	100.0
	D	27.1 [24.7, 29.4]	0.0	99.0
qwen3.8-27b	A	61.2 [60.3, 62.1]	52.6	99.0
	D	14.1 [12.7, 15.6]	0.0	88.5

instead cures a non-compliant clause, reading the clause on its own wrongly flags every such pair (100%), while with override-aware reading the figure is 33.3%.

Comparison with the deployed system. P raises the fewest alarms (14.1%) but misses 80.7% of gaps, since its gap logic covers only the children's-data rules. Overall pair accuracy does not differ significantly between D and P ($p = 0.21$), yet their errors run in opposite directions. D accepts a modest number of extra alarms for complete gap coverage, which we regard as the better trade-off when a missed gap costs more than a false alarm.

B. E2: Behaviour across models

Under A, all three models make the error at the same rate (60.8–61.2%), which places its source in the design of the checker rather than in any particular model. Under D, every model improves (Table IV, Fig. 2b), and none of them misses a gap. McNemar tests give $p < 10^{-168}$ for each model. Under the conservative test, D has a lower false violation rate than A in 18 of 18 (base, model) units (exact sign test, $p = 7.6 \times 10^{-6}$). Qwen reaches the lowest false violation rate but also the lowest violation recall (88.5%), and catches only 38.9% of hidden violations in trap contracts, so the method's gains do not remove model-specific weaknesses.

TABLE V
E3: five independent runs of gpt-oss-20b on 46 contracts (584 pairs). FVR range is the lowest–highest per-run false violation rate (%).

Set-up	All runs agree (%)	Fleiss' κ	FVR range
A	99.3	0.99	60.5–61.1
C	83.4	0.85	45.9–56.6
D	78.4	0.79	22.6–30.8

Agreement is not accuracy. Under A the three models answer 99.3% of pairs identically (Fleiss' $\kappa = 0.99$), largely because they all flag nearly everything; under D they agree on 82.0% ($\kappa = 0.70$), disagreeing on the genuinely subtle pairs. High agreement among models should not be read as evidence of correctness.

C. E3: are the results stable?

We ran A and C/D five independent times on gpt-oss-20b over the 46 contracts of two bases (584 graded pairs): the E2 run restricted to these contracts and four further runs, each with its own empty cache (Table V).

The headline result is stable: in every run D's false violation rate is about half of A's or lower (30.8% against 60.9% in the worst run; mean 26.5% against 60.8%), D misses no gap in any run, and its violation recall stays between 96.9% and 100%. Configuration A gives almost identical answers in every run, again because it flags nearly everything. D varies more ($\kappa = 0.79$; 21.6% of pairs change answer in at least one of the five runs), and its false violation rate moves by up to eight points between runs. The pairs on which the models disagree in E2 are the same ones that a single model is unsure about, which is consistent with the observation that agreement across samples tracks a model's reliability [31]. A deployment should therefore treat findings that are not stable across a re-run as candidates for human review rather than as settled verdicts.

D. E4: re-checking contracts when the law changes

PactScan's regulation-update path embeds a newly added rule, retrieves the related clauses of each existing contract (top-15 by cosine similarity, threshold 0.35), classifies each against the rule, and turns the verdicts into findings. We simulated four DPDP amendments, namely

TABLE VI
E4: re-checking 46 contracts against four simulated DPDP amendments (gpt-oss-120b, 184 checks, %).

Rule	FVR	MGR	F1	Findings/check
Every non-entailing clause	100.0	0.0	50.4	12.15
Contradictions only	31.1	96.2	21.4	0.31
Contract-level (final)	31.1	0.0	76.5	0.58

a 72-hour processor breach-notice duty, erasure within 30 days with written confirmation, a transfer restriction that also requires naming storage locations, and a brand-new duty to name a grievance contact in India. We then re-checked the 46 contracts of two bases (184 checks: 122 NONE, 10 VIOLATES, 52 GAP) under the three rules the system actually went through (Table VI).

The first rule reproduces the incident from development: every compliant contract is flagged, with 12.15 findings per check on average. Restricting findings to contradictions removes 69 points of false alarms but misses 96.2% of gaps, among them every contract that is silent on the new grievance-contact duty. This is the same over-correction that we observed during development. The contract-level rule keeps the lower false alarm rate and recovers every gap (F1 76.5%). Violation recall is 100% under all three rules, and retrieval surfaced the relevant clause for every real violation; the retrieval threshold is permissive (12.8 related clauses per check on average), so errors here come from the verdict and the decision rule, not from retrieval.

The remaining false alarms are instructive. For the erasure amendment, 36 of 40 compliant contracts were flagged because the model marked the processing-details clause, which states that data is deleted within 30 days, as contradicting the amendment for not also requiring written confirmation, even though the contract's deletion clause does require certified deletion. Under the contract-level rule any contradiction takes precedence, so a merely incomplete neighbouring clause outvotes the clause that satisfies the rule: the same silence-as-violation error, moved into the Contradicts label. The other two false alarms are contracts whose deletion or breach clause is cured by a later amendment; this path, unlike Detect, classifies clauses without their overrides.

E. Where the remaining false alarms come from

D's remaining false alarms are not spread evenly (Table VII). Five playbook rules, each stating a single duty (breach-notification window, sub-processor approval, data location, deletion on termination, audit rights), have a false violation rate of 0% on every model. The worst two are compound requirements. DPDP Section 8 bundles security measures, data accuracy and breach notification; the models mark a compliant breach clause as contradicting it because it does not also address data accuracy, which is a duty of the Data Fiduciary rather than of the processor. GDPR Article 8 includes an age threshold;

TABLE VII
False violation rate of D per requirement (%; requirements with 0% on all models omitted).

Requirement	120b	20b	qwen
DPDP s8 (security, accuracy, breach notice)	84.3	100.0	52.8
GDPR Art. 8 (child consent, age threshold)	84.3	65.4	81.9
DPDP s9 / playbook: parental consent	19.7	17.3	17.3
DPDP s4 (grounds for processing)	16.7	65.9	0.0
GDPR Art. 6 (lawful basis)	16.7	31.0	0.0
Playbook: liability cap	15.9	15.9	15.9
Playbook: processing details	1.9	50.9	0.0

the models mark a clause requiring verifiable parental consent for every child as incomplete because it does not state the threshold, although it is stricter than the article. Both are the same silence-as-violation error, now inside a single requirement. Excluding these two requirements, D's false violation rate on gpt-oss-120b would be 8.4%. The liability-cap figure comes from a single base whose liability clause the model judged non-compliant (the base is not the same for every model); because the variants of that base share the clause, the error repeats across them. For every model, the hardest base is the ed-tech contract, which is the only one that actually processes children's data (D: 44.4% on gpt-oss-120b, compared with 10.8–16.2% for the other bases).

VIII. Discussion

Choice of decision unit. The central lesson is that the error comes less from what the model says than from how its answers are aggregated. A clause-level rule asks a question the contract-level answer does not depend on; a better prompt changes the model's words but not the rule's output. Self-reported confidence does not rescue the clause-level rule either. Models can partly estimate their own correctness [32], but in the original rule it was precisely the use of "high-confidence Neutral" as a trigger that turned correctly relabelled silence back into findings. When the decision was moved to the contract and violations were kept attached to the clauses that cause them, most false alarms disappeared and no gaps were lost. PactScan's regulation-update feature showed the same pattern. There, changing what is flagged (every non-entailing clause → contradictions only) simply exchanged one error for another, whereas changing the unit flagged (clause → contract) fixed both.

Decomposing compound requirements. The residual errors point to an inexpensive next step, in which each requirement is expressed as a set of atomic duties (security; accuracy; breach notification) and verdicts are aggregated per duty. Silence on one duty could then no longer contradict another.

Checking contradictions against satisfying clauses. A second residual pattern appears in E4. When a clause addresses only part of a requirement, it is labelled Contradicts for omitting the rest, and the contract-level rule

allows this label to override a clause that does satisfy the requirement. A natural refinement would be to compare each contradiction with the contract's satisfying clauses before it is reported, or, as above, to decompose the requirement.

Recommendation for the deployed system. We recommend that the deployed checker apply its contract-level gap rule to every mandatory requirement, not only to the two children's-data requirements. E1 indicates that doing so closes an 80.7% missed-gap rate at the cost of a few extra alarms.

IX. Threats to Validity and Limitations

Synthetic contracts. The benchmark is synthetic, with six bases in three drafting styles; real DPAs are longer and messier. Its matched design is what makes silence measurable, but results on real contracts may differ.

Labels. Labels were written and reviewed against the requirement texts by the project team, not by practising lawyers, and agreement with a second human annotator has not yet been measured; we release a blind annotation sheet of 120 stratified pairs over 83 anonymised contracts, with a script that reports Cohen's κ [33], so that it can be. As an automated audit (not a substitute for a human annotator), a model used in no experiment (gemini-3.5-flash-lite, temperature 0) labelled the sheet from the same instructions: it agreed on 101 of 120 pairs ($\kappa = 0.73$) and on all 60 None pairs. We checked each of the 19 disagreements against the contract text and found no label error: 12 are model errors (seven explicit violations of a non-mandatory rule called acceptable because the rule is not mandatory; a missing breach-notice or parental-consent clause accepted), two name a violation as a gap, and five follow from the playbook convention above that every DPA must address children's data. Two topics are treated as non-mandatory and GDPR is applied as a second standard regardless of territorial scope; these are stated assumptions.

Verdict step only. Clause types and override links are given, so 36 of the 57 gaps involve a requirement with no clause of its subject left in the contract and are decided without the model; the remaining 21 gaps were decided from model verdicts, and all 21 were caught. End-to-end accuracy with automatic extraction is not measured.

Dependence. Variants share clauses and each distinct question is answered once per run, so pair-level tests overstate certainty; the per-base sign test is our conservative check.

Models and settings. Three open models on one provider, at a single temperature; free-tier quotas limited E3 and E4 to two bases and moved E3 to gpt-oss-20b. E4's amendments are simulated, and its re-check path classifies clauses without their overrides.

X. Conclusion

Clause-by-clause LLM compliance checking mistakes silence for violation: on a matched benchmark of DPDP vendor DPAs, about 61% of pairs on which nothing is wrong were flagged by every model tested. A conservative prompt changes the model's verdicts but not this outcome; deciding at the level of the contract, with subject routing and override-aware reading, cuts false violations to 14–27% across models, misses no mandatory gap and keeps violation recall at 88–99%, and the gain repeats across independent runs, although about one pair in five changes answer in at least one of five runs. When contracts are re-checked against new regulations, the same contract-level rule avoids both the flood of false alarms and the hidden gaps that the two earlier rules produced. The remaining errors point to compound requirements, and to incomplete clauses read as contradictions, as the next targets. Beyond the verdict step, three questions about the same pipeline remain open: whether an AI-drafted redline, once approved, actually resolves the finding it was written for when the contract is re-checked; whether the model's stated confidence predicts which verdicts are wrong; and how much of the remaining disagreement between models disappears under a larger, lawyer-labelled benchmark. We release the benchmark, harness and raw verdicts to support further work on compliance checking under the DPDP Act.

Data and Code Availability

The benchmark, the experiment harness, the raw model verdicts and the scripts that produce every number in this paper are in the research/ folder of the project repository, https://github.com/saurabh-1702/PactScan-AI (branch research/silence-not-violation).

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